Gated Qwen-Drive with Camera-Based Emergency Braking: AlpaSim E2E Closed Loop Challenge Technical Report

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Abstract

This report describes our Physical AI AV (PAI-AV) track entry. The learned policy pairs the driving-pretrained Qwen-Drive-1.0-4B vision-language model [4] with a layer-wise flow-matching expert that outputs 2.5 s of acceleration and curvature controls, plus a soft visual mask on the expert's cross-attention. Against an identically trained baseline, open-loop error is about equal overall and lower on left turns. In local closed-loop evaluation on the same 150 scenes, two rollouts each, the mask raises the mean scene score from 0.479 to 0.517 (paired difference $+0.038$, 95% bootstrap CI $[-0.018, +0.093]$, wins/losses 37/26, sign test $p = 0.21$). A rule-based emergency brake (AEB) added at deployment raises it further to 0.599 ($+0.082$, CI $[+0.023, +0.143]$), and mean at-fault collisions fall from 34.0 to 11.0 scenes.

1 Introduction

A small planning error changes what a driving policy sees next, an effect that only closed-loop evaluation reveals. The AlpaSim challenge scores policies in closed loop on held-out scenes [17]: a scene scores zero on an at-fault collision, leaving the road or leaving the recorded corridor. In our early runs, at-fault collisions were the largest source of loss, mostly the ego driving into slower or stopped vehicles, in scenes where its plan did not brake. We report two responses: a learned mask that limits which image evidence the action expert sees and is trained so that navigation intent can select that evidence, and a detector-based brake that acts when the policy still fails.

2 Related work

Vision-language-action driving. Qwen-Drive-1.0 [4] provides a vision-language foundation for driving with a planning expert, Alpamayo-R1 [10] pairs a reasoning model with a diffusion trajectory decoder, and DriveMoE [7] adds expert routing to Drive-π_0, which adapts π_0 [6]. We train a π_0-style expert on Qwen-Drive's backbone. **Limiting visual evidence.** S2 [5] learns per-token soft masks under a visual evidence budget, with an ungated path. We reuse the gate head and budget idea, but act on the expert's cross-attention, fix the budget by construction and train the mask by distillation (Section 4). **Copycat behavior.** Imitation policies can latch onto their own past motion [1]; ChauffeurNet [2] drops past motion in training. We tried history dropout and temporal ensembling [3], without clear benefit (Appendix H).

3 Method summary

The driver (Fig. 1) has an end-to-end policy that replans every 0.5 s and a detector-based AEB that can only slow the plan along its own path. We submitted three variants of one checkpoint: AEB at 10 Hz, AEB at 5 Hz, and no AEB. Code builds on starVLA [18] and py123d_garage [19]; evaluation uses AlpaSim [17].

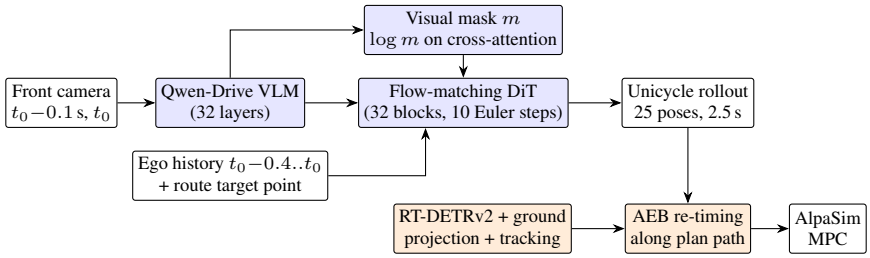


Figure 1: System overview. Blue: learned, trained end to end. Orange: pretrained detector and hand-written rules, added only at deployment.

4 Policy description

Inputs. Two front wide camera frames (current, pixel budget 921,600; $t_0-0.1$ s, budget 174,080) and the fixed prompt "predict trajectory" enter the VLM. Ego state and navigation go only to the expert: five 10 Hz steps of $[x, y, \text{heading}, \text{speed}, a_{\text{lon}}]$ with a target point $(x_{\text{tp}}, y_{\text{tp}})$ repeated per step and a presence flag (state dimension 8). In training the target point is the first future ego position at least 25 m away. At deployment it is the first point of AlpaSim's map route at least 25 m away in a straight line; because the route starts about 40 m ahead of the ego, the gap is bridged with a cubic Hermite curve.

Expert and outputs. The expert follows the π_0-style layer-wise design [6, 7]: a 32-block DiT [8] alternates self-attention with cross-attention, each cross-attention block reading a different VLM layer after a learned projection. It is trained by flow matching [9] on velocity MSE, with time drawn from a shifted Beta(1.5, 1) and $s = 0.999$, and sampled with 10 Euler steps at inference. It predicts 25 steps of [acceleration, curvature] at 10 Hz, z-scored per dimension, using the unicycle action space vendored from NVIDIA Alpamayo 1.5 [10, 11]. Integrating from the current speed gives the poses that are served.

Baseline. The baseline has the same architecture and training, with the target point dropped with $p = 0.5$ per training sample.

Visual mask. A gate reads each visual token's final-layer VLM hidden h_p (detached) and a query q embedding the scaled target point and ego state, and computes a score ℓ_p from $[h_p, q, h_p \odot q]$. This head form follows the visual evidence budget of S2 [5]. Scores are standardized per frame to z_p, and $m_p = \sigma(b + 2z_p)$, with b found by bisection so that the frame mean of m equals the budget $\rho = 0.2$ exactly; no budget loss is needed. Unlike S2, which multiplies token embeddings by a floored mask inside the executor, we add $\log m_p$ to the expert's cross-attention logits and leave text tokens and the VLM unmasked. Training uses three paths from one VLM forward with shared noise and time:

- **A** (ungated, target dropped with $p = 0.5$): flow-matching loss, updates VLM and DiT.
- **B** (mask held fixed, target given): $\|v_B - \text{sg}(v_A)\|^2$, updates the DiT; this is the inference path.
- **C** (mask, no target in the DiT state, gate still sees it): $\|v_C - \text{sg}(v_A)\|^2$, updates the gate only.

B and C are distilled only on samples where path A had the target, so the gate must make the target's information visible through the image evidence it keeps.

Emergency brake. An RT-DETRv2-R18 [12, 13] trained on COCO [14] runs on the front frame at 640×640 and keeps person, bicycle, car, motorcycle, bus, train and truck boxes with score above

0.5. The central 40% of each box's bottom edge is cast through the announced f-theta calibration onto the flat ground plane of the rig. Among obstacles within 1.3 m laterally of the planned path (extended straight to 80 m), the nearest gives the gap d from the front bumper, measured along the path. The closing speed c is a line fit over the last 1 s of d, using at least 6 samples (3 for the 5 Hz variant); a jump above 4 m restarts the track. With

$$g = d - 2.5\,\mathrm{m} - 0.4\,\mathrm{s} \cdot c, \qquad a_{\mathrm{req}} = \frac{c^2}{2\max(g,\, 0.1\,\mathrm{m})}, \tag{1}$$

the plan is limited to a ramp from the current speed down to the lead speed $\max(v - c, 0)$ at $\mathrm{clip}(1.2\,a_{\mathrm{req}}, 3, 8)$ m/s^2 when $g \le 0$ or $a_{\mathrm{req}} > 3$ m/s^2. If only $d/c < 3$ s, the plan may not exceed the current speed. The plan is re-timed along its own path from the ego's projection onto it, so a braked ego does not accelerate to catch up with a stale plan; on release, speed is held for one tick and the policy replans.

5 Resources and training

Data. NVIDIA PhysicalAI-Autonomous-Vehicles [16] (NVIDIA Autonomous Vehicle Dataset License Agreement): the front wide camera and egomotion of its train split. No LiDAR, radar or machine labels were used.

Pretrained weights. Qwen-Drive-1.0-4B [4] for the VLM (Apache-2.0), and RT-DETRv2-R18 (Apache-2.0, `PekingU/rtdetr_v2_r18vd`) for the detector. **Private resources:** none.

Training. Both runs (W&B `ccb412c9` baseline, `03bdfc50` mask) used the same commit, seed 42 and 12 epochs (38,400 steps) on 12 Aurora XPU tiles, with the VLM frozen in the first epoch, in the spirit of LP-FT [15]. Settings and curves are in the appendix.

Open-loop results. Table 1 uses a fixed set of 240 validation samples (80 each of left, right and straight). Overall error is about equal: the mask is better on left turns (FDE 1.100 vs. 1.213, miss rate 0.150 vs. 0.200), slightly better on right-turn FDE, and worse when driving straight. Table 2 gives the controls for the mask run. Removing the images raises FDE more than removing the target point, and the images matter in both runs (baseline with black frames: ADE 0.441, FDE 1.408). At the end of training, path C reached a loss of 0.0122, against 0.0160 with shuffled mask values and 0.0156 with no mask.

Table 1: Open-loop error at step 38,400, baseline / mask. ADE and FDE in m over 2.5 s; miss: any waypoint $>$ 2 m.

	ADE	FDE	Miss rate
All	0.337 / 0.341	1.029 / 1.023	0.125 / 0.125
Left	0.405 / 0.376	1.213 / 1.100	0.200 / 0.150
Right	0.367 / 0.360	1.136 / 1.089	0.087 / 0.100
Straight	0.239 / 0.287	0.737 / 0.879	0.087 / 0.125

Table 2: Mask-run controls, all samples.

Control	ADE	FDE
No mask	0.339	1.016
Shuffled mask	0.344	1.031
Mirrored target in gate	0.348	1.049
No target in DiT, no mask	0.367	1.135
No target in DiT, mask	0.359	1.094
All image tokens removed	0.456	1.418
Black input frames	0.486	1.564

Closed-loop results. Table 3 uses the same 150 scenes in every row, from a subset of the 441-scene curated validation split, with the `dev` challenge preset (40 m route offset). A scene scores 0 on an at-fault collision, offroad or lateral corridor exit, and otherwise $\min(\mathrm{progress}/0.8, 1)$. First and second rollouts ran on different hardware (Blackwell and Ada hosts), so differences between them mix host and run-to-run variation. Per-scene means of two rollouts give: mask vs. baseline, $+0.038$, 95% CI $[-0.018, +0.093]$, wins/losses 37/26, sign test $p = 0.21$; AEB at 10 Hz vs. mask, $+0.082$, CI $[+0.023, +0.143]$, 41/28, $p = 0.15$. AEB at 5 Hz vs. 10 Hz: -0.009, CI $[-0.065, +0.044]$, 34/34, $p = 1.00$. The appendix gives every comparison.

6 Runtime and reproducibility

Serving. The image is self-contained, runs in the challenge sandbox and starts without arguments. The policy replans every 0.5 s and serves the cached plan on the 10 Hz grid in between.

Table 3: Local closed-loop results on the same 150 scenes. Failures in scenes; rear-ended scenes are not at fault and only stop progress accrual. Mean Drive includes video recording.

Configuration	Host	Score	At-fault coll.	Offroad	Corridor exit	Rear-ended	Mean Drive (s)
Baseline, 1st	Blackwell	0.476	42	25	8	12	0.133
Baseline, 2nd	Ada	0.481	41	19	15	9	0.066
Mask, 1st	Blackwell	0.499	35	22	12	15	0.110
Mask, 2nd	Ada	0.534	33	23	8	15	0.146
Mask + AEB 10 Hz, 1st	Blackwell	0.607	12	27	11	25	0.201
Mask + AEB 10 Hz, 2nd	Ada	0.591	10	31	13	23	0.085
Mask + AEB 5 Hz	Ada	0.590	13	27	10	28	0.264

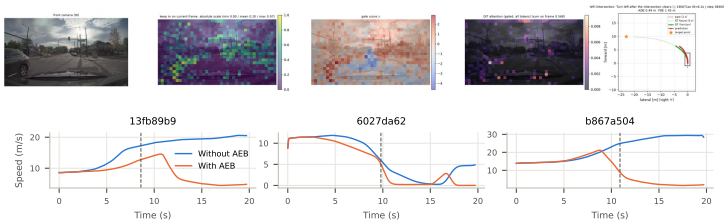


Figure 2: Top: mask-run validation sample, left turn (current frame; keep value; gate score; DiT attention; bird's-eye view). Bottom: ego speed in three scenes without and with AEB (dashed: collision without AEB).

Latency and memory. Drive times in Table 3 are local means with video recording and up to 12 rollouts per driver on shared hosts, so they are only indicative. With one rollout per replica and no recording (RTX 6000 Ada, shared GPU), the mean Drive call was 0.10–0.11 s for both AEB variants (Appendix I). The detector allocates about 0.2 GiB on top of the policy's roughly 11.6 GiB.

Reproducibility. The deployed configuration matches the W&B runs. Code: `https://github.com/Azuma413/VLA4AD_Research`, branch `dev/alpasim`, commit `80ab247`.

7 Limitations and safety considerations

The AEB's evidence is limited. Its thresholds were tuned by replaying detections offline on the same 150 scenes used in Table 3, so the gain may be optimistic for the private set.

Remaining collisions. Of the 12 at-fault collisions in the first AEB rollout (our judgment from logs, replay and video; Appendix G), 6 involved an obstacle about 2 m to the side that entered the corridor too late or never, a result of narrowing it to suppress false braking; 2 came from the side or behind-left; 2 were close-range contacts where the estimated distance stopped changing; and in 2 braking began too late or did not fully stop the ego.

Side effects. In the first rollouts, rear-end collisions increase from 15 to 25 scenes, which is not at fault here but would be a real safety concern on a road, and 3 previously clean scenes newly collide.

Lateral failures dominate. In the first AEB rollout, offroad and corridor exits make up 0.253 of the 0.393 total score loss, and the AEB cannot address them. Offroad events occur a median of only 0.67 m from the route, and clean rollouts deviate comparably, so we did not deploy a route-distance rule.

Reliance on ego motion. Without the AEB, first-incident front collisions occur in 25.7% of mask rollouts (77/300) against 29.9% for the baseline (90/301). Among them, the ego gained more than 2 m/s in the last 3 s in 30% vs. 42% of cases, and the last plan did not slow down in 64% vs. 68%. The mask appears to reduce this failure, but most front collisions still occur with no braking in the plan.

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Table 4: Training configuration (identical for both runs unless stated). Source: W&B config of `ccb412c9` and `03bdfc50`; the bundles deployed in the images carry the same values.

Item	Value
Epochs / steps	12 / 38,400 (3,200 steps per epoch)
Hardware	1 Aurora node, 12 XPU tiles, one rank per tile (batch size 4 per tile)
Precision	bf16; gradient checkpointing on; max gradient norm 1
Optimizer	AdamW, betas $(0.9, 0.95)$, $\epsilon = 10^{-8}$, weight decay 0
Learning rates	VLM 10^{-6}; expert 5×10^{-5}; gate 10^{-4} (mask run)
Schedule	200 warmup steps, cosine to 10^{-8}; VLM frozen in epoch 1, then warmed up and decayed over the rest
Repeated diffusion steps	4 noise/time draws per image
Prompt	"predict trajectory" (no target or speed in the prompt)
Images	Front wide camera; pixel budgets 921,600 (t_0) and 174,080 ($t_0 - 0.1$ s); token per 32×32 pixels
VLM	`Qwen/Qwen-Drive-1.0-4B`: 32 layers, hidden 2560, SDPA attention
Expert	LayerwiseFM DiT: 32 blocks interleaving cross- and self-attention, width 1024, 16 heads of dim 64, dropout 0.2, AdaNorm
State	Horizon 5, dim 8: x, y, heading, speed, a_{lon}, x_{tp}, y_{tp}, flag; per-dim q01/q99 to $[-1, 1]$
Action	Horizon 25, dim 2: acceleration, curvature; per-dim z-score
Flow matching	Beta(1.5, 1) time, $s = 0.999$; velocity MSE; 10 Euler steps at inference; 1000 time buckets
Target dropout	$p = 0.5$ per sample (path A in the mask run); columns zeroed, flag 0
Eval	Every 250 steps on 240 fixed validation samples; open loop
Mask (mask run only)	Scheme three_path; parameterization budgeted; $\rho = 0.2$; contrast 2; head hidden 256
Gate query	Target point $/25$ and flattened normalized ego state without target columns
Gate input	Final-layer VLM hidden of visual tokens, detached, standardized per channel; RMSNorm

Table 5: Deployment configuration. Replanning, CUDA graph and detector settings are defaults or baked overrides of the submitted images.

Item	Value
Replanning	Every 5th 100 ms interval (0.5 s); plan cached in the local frame and re-sliced each tick
Temporal ensembling; action samples	Off (1 plan; 1 sample) in all three submissions
Linear-attention kernels	`auto`: flash-linear-attention Triton kernels, falling back to the reference implementation
Expert denoising	Replayed from a CUDA graph
Target point	25 m, on the route; route resampled at 0.1 m and extended 2 m past the farthest target
Controller gains	Official defaults (no `-controller-gains`)
Detector	RT-DETRv2-R18, COCO, 640×640 on the front wide frame, score > 0.5, no NMS (DETR queries)
Classes	person, bicycle, car, motorcycle, bus, train, truck
Ground projection	7 points on the central 40% of the box bottom; flat ground $z = 0$ in the rig frame; range < 80 m
Corridor	1.3 m half width around the plan path; path extended straight to 80 m
Front overhang	4.05 m, rear axle to front bumper; hits behind -1 m of the bumper ignored
Track	1 s window; at least 6 samples (10 Hz) or 3 (5 Hz); reset on a jump above 4 m or no detection
Stop gap; latency	2.5 m; 0.4 s
Brake trigger; deceleration	$a_{\mathrm{req}} > 3$ m/s^2 or $g \leq 0$; $\mathrm{clip}(1.2\, a_{\mathrm{req}}, 3, 8)$ m/s^2
No-acceleration condition	$d/c < 3$ s
Re-timed plan grid	0.1 s
Detection rate	Every front frame (10 Hz variant); at least 0.2 s apart (5 Hz variant)

A Full configuration

B Data and action representation

Sampling. Clips with valid egomotion and front camera in the release's train split are used. Each dataset item is one clip with t_0 drawn uniformly from its valid range, so an epoch visits each clip once at a random time. Only the dense part of the egomotion, about 20 s around the camera window, is read. The two input frames are the front wide frames nearest t_0 and $t_0 - 0.1$ s.

State. Per step $[x, y, \mathrm{heading}_{\mathrm{rel}}, \mathrm{speed}, a_{\mathrm{lon}}]$ at $t_0 - 0.4, \ldots, t_0$ in the ego frame at t_0. These are quantities AlpaSim also provides. The target point in the ego frame at t_0 is repeated per step, followed by a presence flag.

Target point. The first future position at least 25 m from the current one in a straight line, or the clip's last recorded position when it ends earlier. In the deployed model's training, the latter case was not removed (a later experiment, Appendix H, drops it because such a target marks where the ego stops, which a route target never does).

Action. 25 per-step [acceleration, curvature] pairs from t_0, obtained with Alpamayo 1.5's regularized least-squares fit (smoothing heading, velocity, acceleration and curvature; curvature is 0 below 0.6 m/s) and integrated trapezoidally from the current speed v_0 to positions at $t_0 + 0.1, \ldots, t_0 + 2.5$ s.

Statistics and validation. Normalization statistics come from 6,000 train clips $\times$ 8 time steps. The 240 validation samples are 80 per maneuver, classified by the heading change over the next 3 s with a $\pm 15^\circ$ threshold.

C Mask details and diagnostics

Budgeted mask. For the token scores ℓ_p of one frame, $z_p = (\ell_p - \bar{\ell})/(\mathrm{std}(\ell) + 10^{-6})$ and $m_p = \sigma(b + \alpha z_p)$ with $\alpha = 2$, where 40 bisection steps on $b \in [-30, 30]$ give $\mathrm{mean}_p\, m_p = \rho$. The bisection is not differentiated. Instead, since $\sum_p \sigma(b + \alpha z_p) = N\rho$ defines $b(z)$ implicitly with $\partial b/\partial z_q = -\alpha w_q$, $w = \sigma'/\sum \sigma'$, a zero-valued term $-\alpha \sum_q w_q(z_q - \mathrm{sg}[z_q])$ is added to supply exactly this gradient. A uniform mask is impossible by construction and the spread of m is set by α, so the mask cannot collapse to all-keep or binarize.

Where it acts. The expert's cross-attention logits receive $\log m_p$ on visual tokens, 0 on text tokens and $-\infty$ on padding, in every cross-attention block. The VLM itself is not masked. The mask applies to both frames with a budget per frame, from one gate head.

Gradients. Path A updates VLM and DiT. Path B uses a detached mask and detached VLM features and updates the DiT. Path C uses detached VLM features; the trainer applies its gradient to the gate only. The total action loss is $L_A + L_B$; L_C is backpropagated separately. All paths share noise and time, and B and C are weighted by whether the sample's path A had the target.

Controls. *No mask*: same inputs, no $\log m$. *Shuffled*: per frame, the same keep values at random positions. *Mirrored*: the mask the gate gives for the target with $y \to -y$. *No image tokens*: keep 0 on every visual token. *Black frames*: all frames replaced by black images and the VLM re-run, with no mask. *No target*: flag 0, target columns zeroed in the DiT state; with the mask, the target reaches the policy only through the gate.

Table 6: Mask diagnostics over the validation set (current frame), step 38,400.

Diagnostic	Value
Mean keep, history frame	0.200
Mean keep, current frame	0.200
Spearman between masks of two samples (1 = one fixed pattern)	0.055
Mask variance explained by the mean row pattern	0.024
Mask variance explained by the mean column pattern	0.003
Spearman between a sample's mask and its DiT attention	0.827
DiT cross-attention mass on text tokens	0.261
Keep on image left minus right half, left turns	0.002
Same, right turns	0.025
Same, straight	0.011

Table 7: Mask-run training signals: mean of the first 50 and last 200 logged points (one point per 10 steps).

Signal	First 50	Last 200
Path A (flow matching)	0.9586	0.1790
Path B (distillation, DiT)	0.0521	0.0071
Path C (distillation, gate)	0.0674	0.0122
Path C, shuffled mask (no gradient)	0.0688	0.0160
Path C, no mask (no gradient)	0.0680	0.0156
Mean keep, current frame	0.2000	0.2000
Mean keep, history frame	0.2000	0.2000
Std of gate scores	2.0426	3.4981
Fraction with target (path A)	0.4987	0.5026
Gradient norm	24.5550	4.1008
Samples per second	20.9434	6.6069
Step time (s)	2.2586	7.1135

Table 8: Baseline training signals, as in Table 7.

Signal	First 50	Last 200
Flow-matching loss	0.9424	0.1796
Loss, acceleration	0.9591	0.2635
Loss, curvature	0.9256	0.0957
Fraction with target	0.4944	0.4997
Gradient norm	24.6998	3.7741
Samples per second	28.5946	6.9585

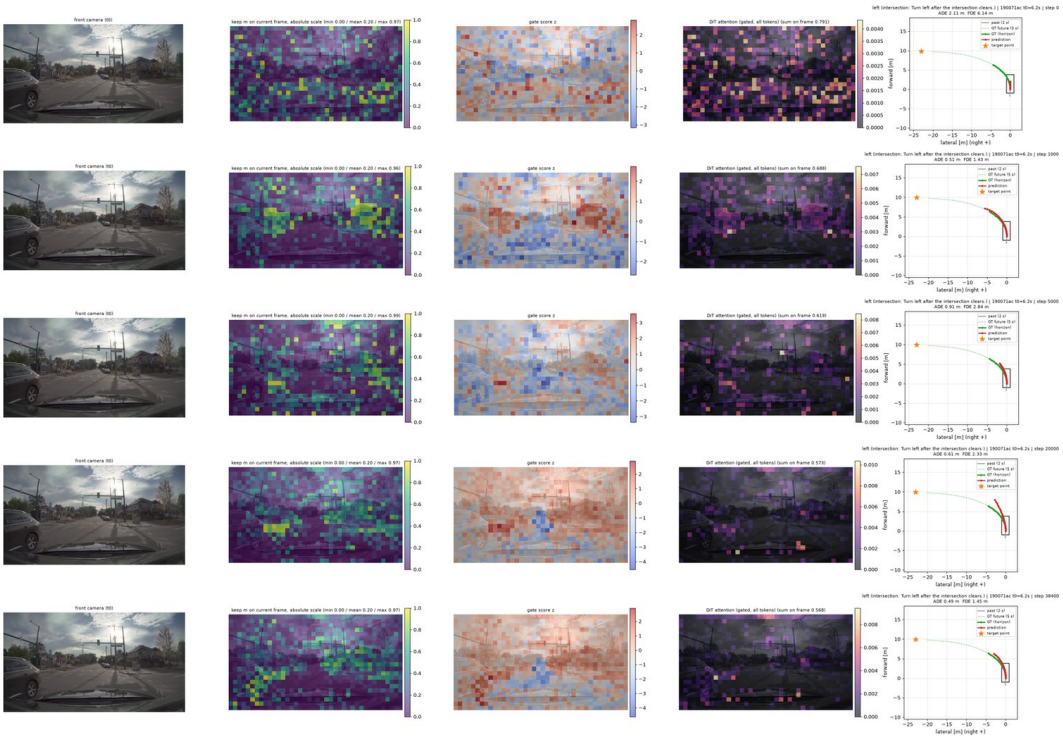


Figure 3: Mask on one validation sample across training: steps 0, 1,000, 5,000, 20,000, 38,400, top to bottom. Panels as logged, left to right: current front frame; keep value m; standardized gate score z; DiT attention on the frame; bird's-eye view with past, ground truth, prediction and target point (lateral axis positive to the right).

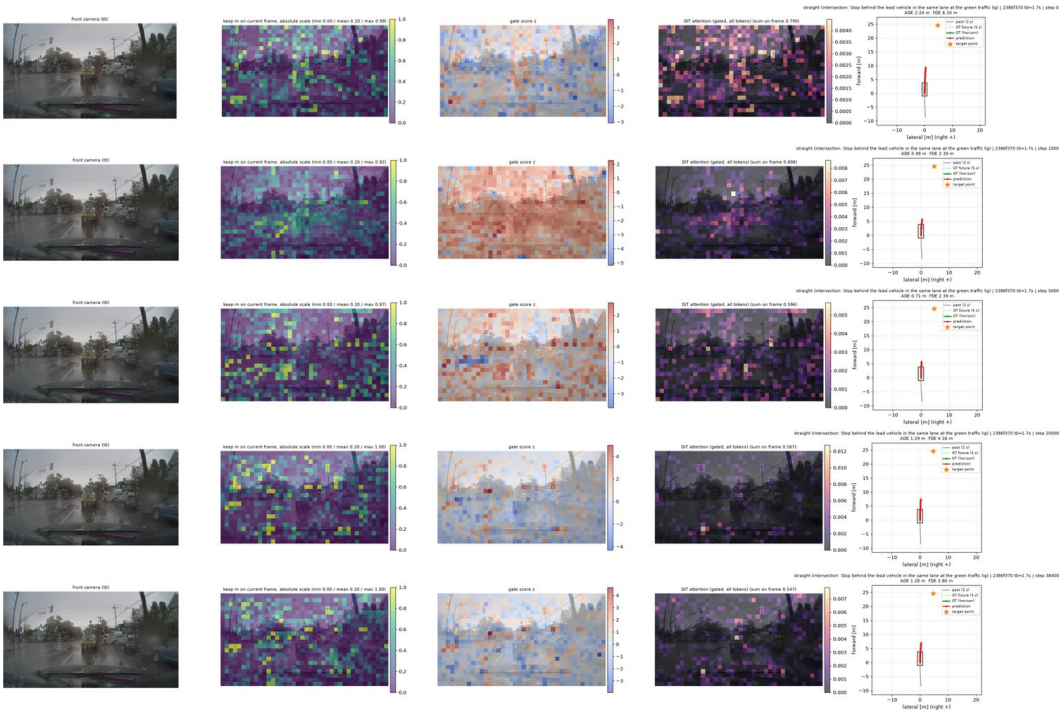


Figure 4: As Fig. 3, for a straight sample behind a lead vehicle.

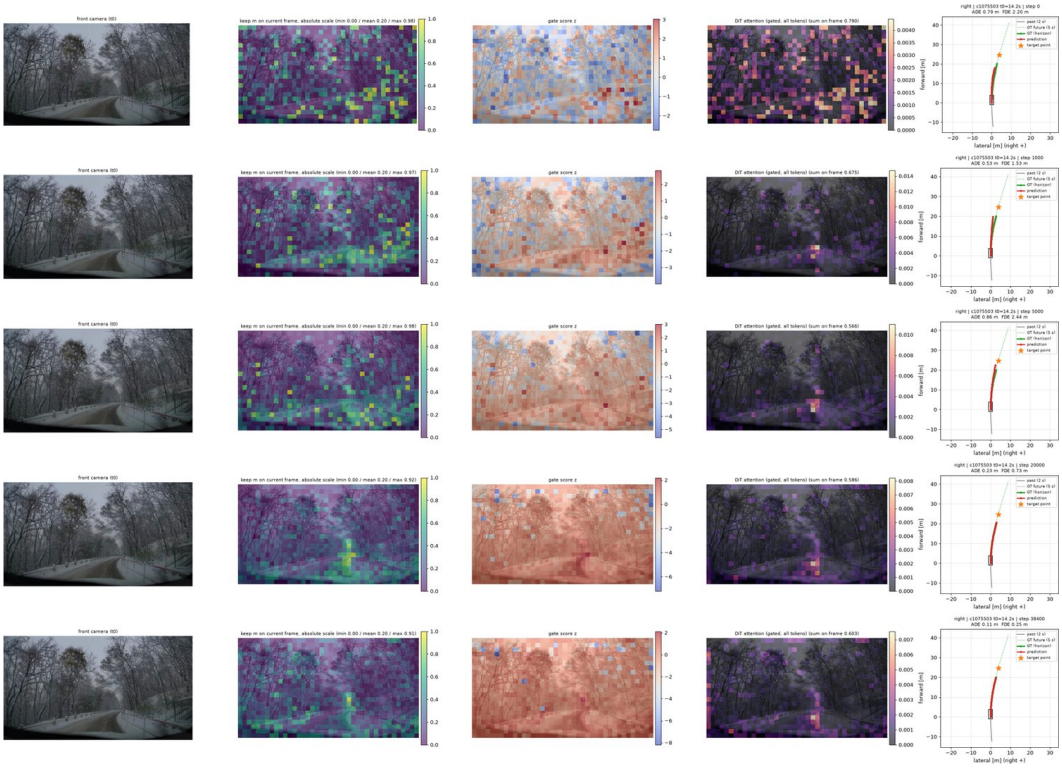


Figure 5: As Fig. 3, for a right-turn sample.

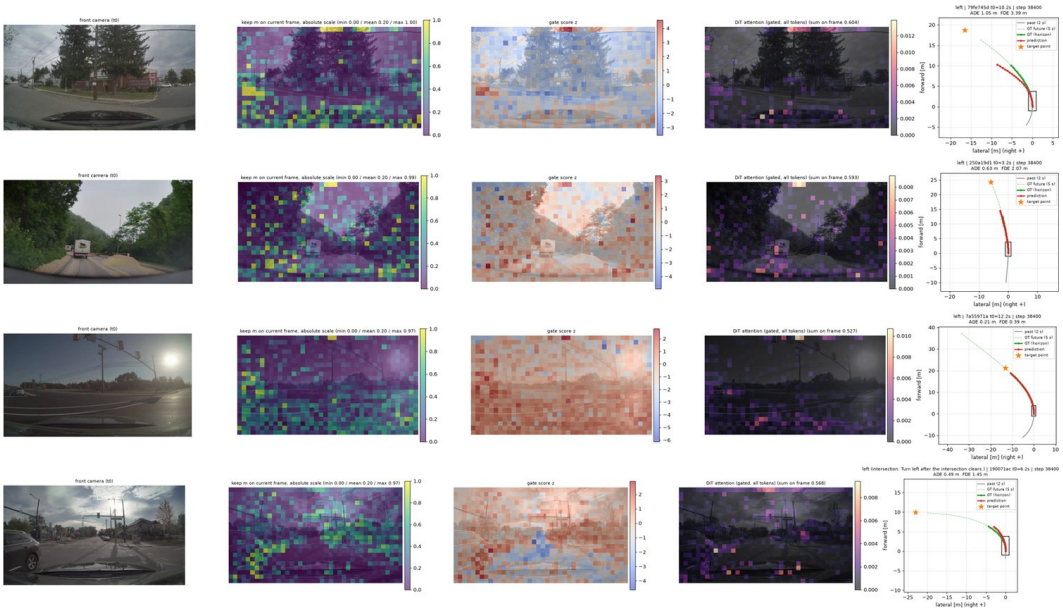


Figure 6: Mask-run left-turn samples at step 38,400. Panels as logged, left to right: current front frame; keep value m; standardized gate score z; DiT attention on the frame; bird's-eye view with past, ground truth, prediction and target point (lateral axis positive to the right).

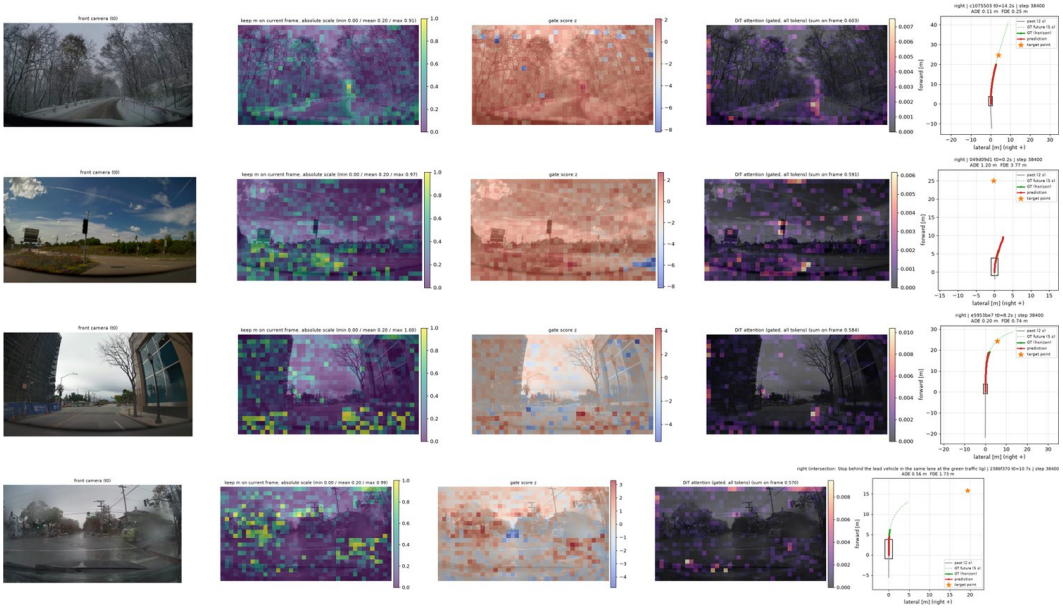


Figure 7: Mask-run right-turn samples at step 38,400, panels as in Fig. 6.

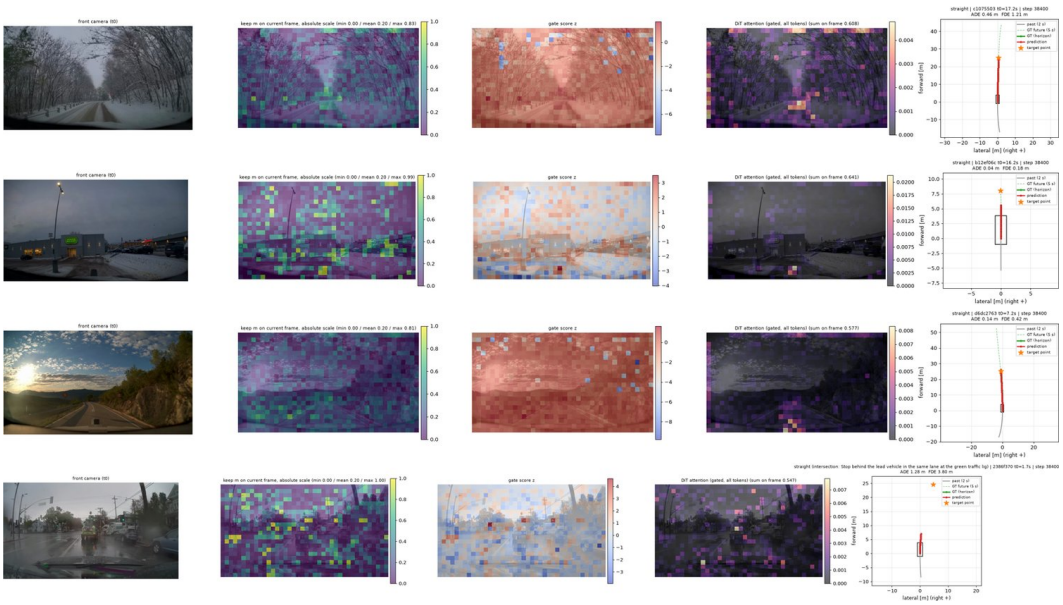


Figure 8: Mask-run straight samples at step 38,400, panels as in Fig. 6.

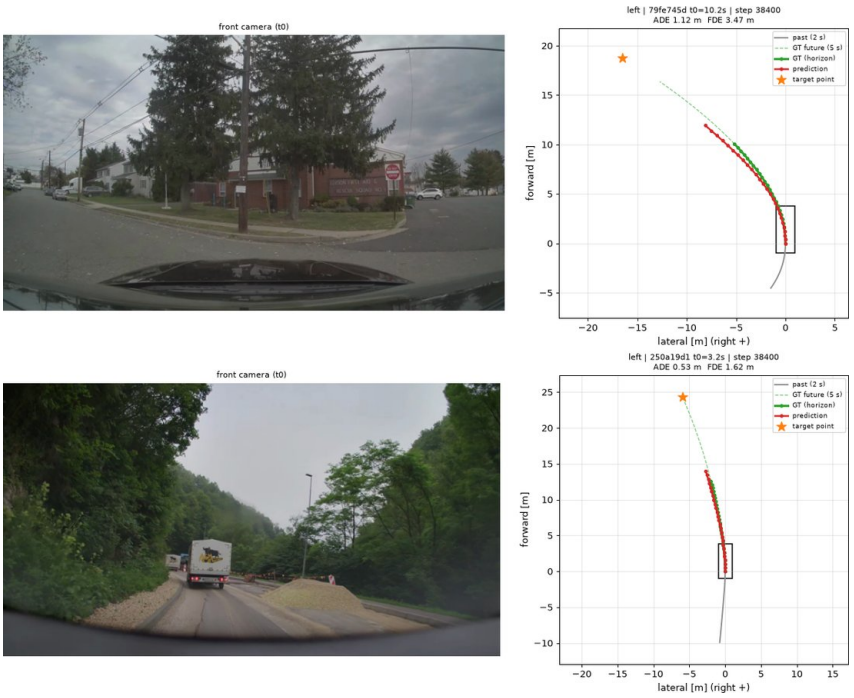


Figure 9: Baseline on the first two left-turn samples of Fig. 6, step 38,400 (front frame and bird's-eye view; the baseline has no mask).

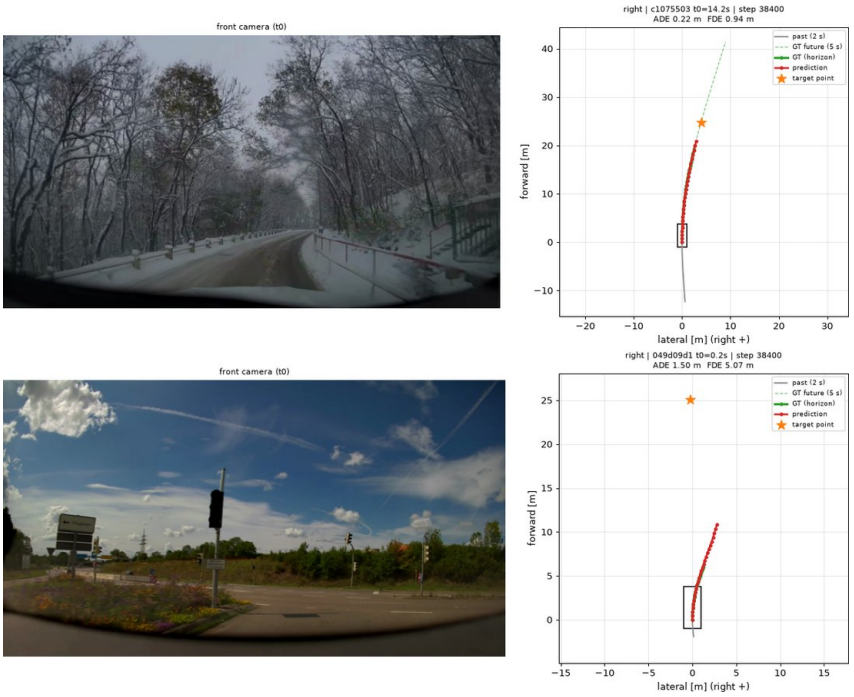


Figure 10: Baseline on the first two right-turn samples of Fig. 7.

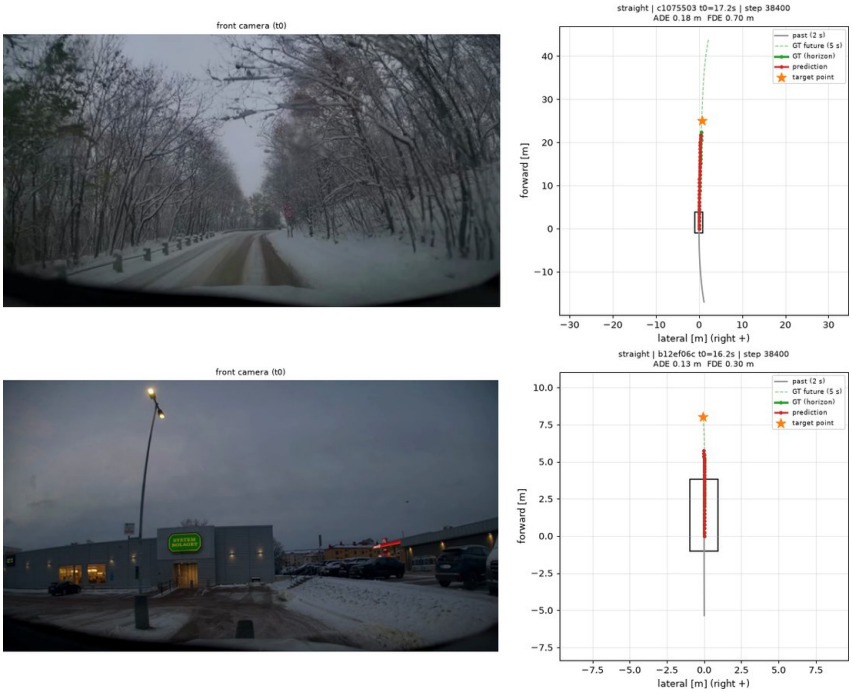


Figure 11: Baseline on the first two straight samples of Fig. 8.

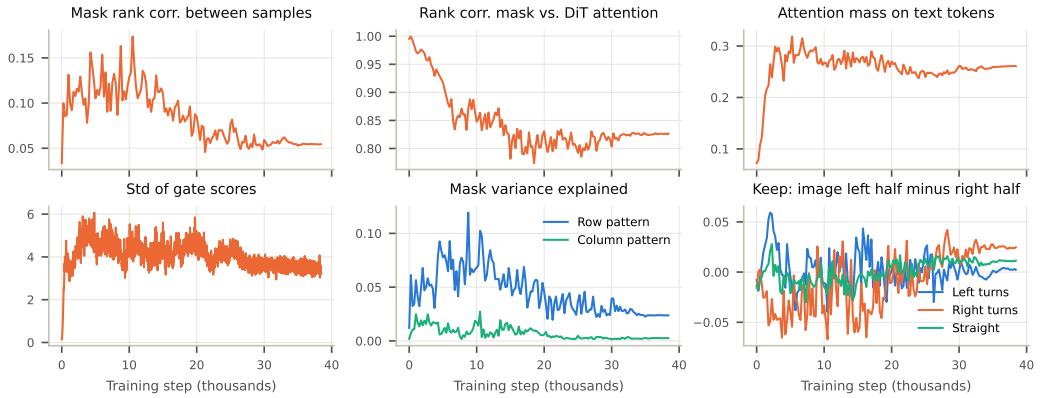


Figure 12: Mask diagnostics over training. The mask's rank correlation with DiT attention falls from near 1 to about 0.83, attention on text tokens rises early, and row and column patterns explain little of the mask's variance.

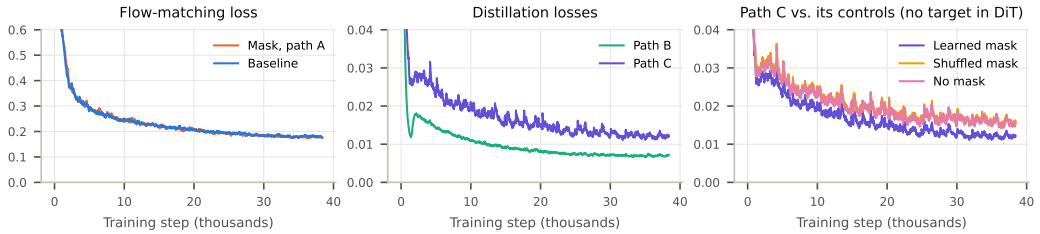


Figure 13: Training losses, smoothed (EMA 0.05). Baseline and path A match. Learned-mask path C stays below its shuffled and no-mask controls.

D Open-loop results in full

Both runs use the final checkpoint. The best evaluation over training is lower than the final one, on this fixed set of 240 samples: baseline ADE 0.302 at step 32,250 (final 0.337), mask 0.324 at step 34,750 (final 0.341). We did not select on this set. Each evaluation draws one flow-matching sample per input, so these values carry sampling noise; no confidence intervals were computed.

Table 9: Open-loop metrics at step 38,400 on the 240 fixed validation samples (m; miss rate as a fraction). L2 at the stated horizon in s; Lat./Lon.: absolute final error.

	Run	ADE	FDE	L2@0.5	L2@1.0	L2@1.5	L2@2.0	L2@2.5	Lat.	Lon.	Miss
All	Baseline	0.337	1.029	0.025	0.117	0.300	0.574	0.943	0.478	0.802	0.125
	Mask	0.341	1.023	0.028	0.126	0.308	0.578	0.939	0.457	0.796	0.125
Left	Baseline	0.405	1.213	0.034	0.151	0.365	0.686	1.115	0.768	0.790	0.200
	Mask	0.376	1.100	0.034	0.146	0.347	0.637	1.014	0.667	0.712	0.150
Right	Baseline	0.367	1.136	0.025	0.117	0.320	0.633	1.041	0.516	0.931	0.087
	Mask	0.360	1.089	0.031	0.136	0.321	0.606	0.997	0.526	0.867	0.100
Straight	Baseline	0.239	0.737	0.017	0.083	0.215	0.404	0.672	0.149	0.686	0.087
	Mask	0.287	0.879	0.020	0.097	0.255	0.490	0.806	0.179	0.810	0.125

E Driver pipeline

Observation handling. The driver answers AlpaSim's gRPC API. JPEGs of the mapped cameras are decoded on arrival; only the front wide camera feeds the policy and the detector. Calibration comes from the session's announced f-theta intrinsics, scaled to the decoded frame, with the rig file

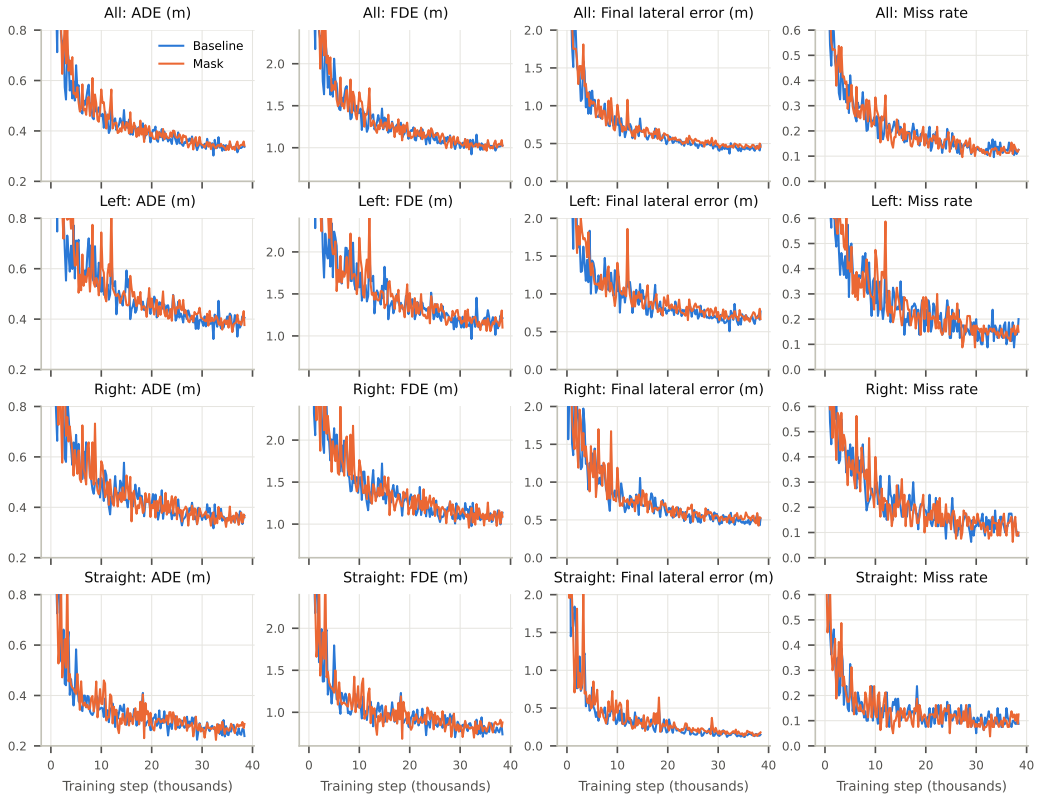


Figure 14: Validation metrics over training, every 250 steps. Vertical axes are clipped, so the first evaluations are off scale.

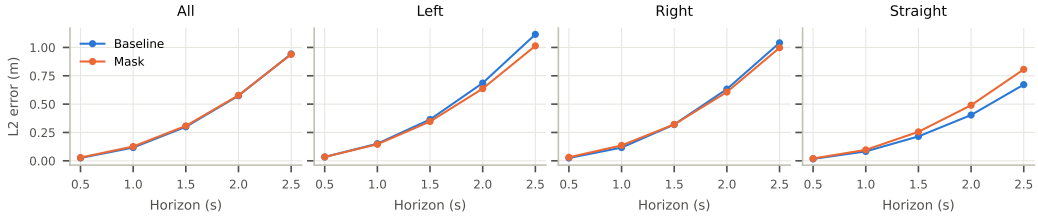


Figure 15: L2 error along the 2.5 s horizon at step 38,400.

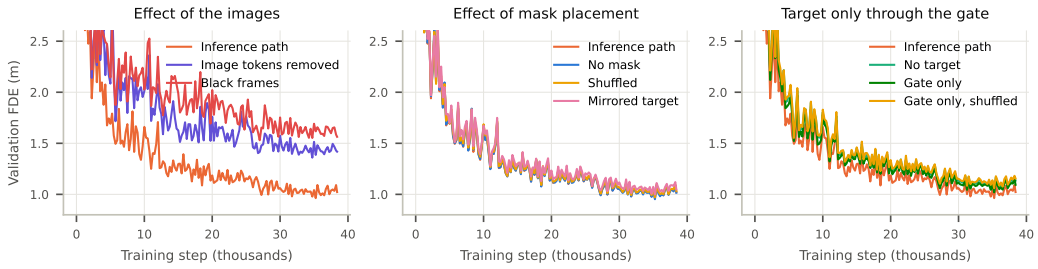


Figure 16: Mask-run evaluation controls over training (validation FDE). Definitions in Appendix C.

as a fallback. Egomotion is stamped on the newest observation bundle at every drive call, so the 0.5 s replanning still reads a full 10 Hz history.

Table 10: Evaluation controls of the mask run at step 38,400, with the noise of the main prediction. Each cell group is ADE, FDE, miss rate.

	All			Left			Right			Straight		
Inference path	0.341	1.023	0.125	0.376	1.100	0.150	0.360	1.089	0.100	0.287	0.879	0.125
No mask	0.339	1.016	0.129	0.376	1.096	0.138	0.358	1.086	0.125	0.282	0.866	0.125
Shuffled mask	0.344	1.031	0.133	0.382	1.115	0.150	0.363	1.099	0.125	0.287	0.879	0.125
Mirrored target in gate	0.348	1.049	0.129	0.392	1.154	0.175	0.363	1.114	0.087	0.288	0.881	0.125
No target in DiT, no mask	0.367	1.135	0.167	0.395	1.217	0.188	0.403	1.261	0.163	0.303	0.926	0.150
No target in DiT, mask	0.359	1.094	0.142	0.383	1.150	0.163	0.391	1.210	0.125	0.302	0.921	0.138
No target in DiT, shuffled mask	0.374	1.155	0.163	0.406	1.249	0.175	0.411	1.285	0.175	0.305	0.932	0.138
All image tokens removed	0.456	1.418	0.242	0.469	1.449	0.263	0.507	1.601	0.275	0.393	1.204	0.188
Black input frames	0.486	1.564	0.304	0.518	1.598	0.362	0.561	1.882	0.375	0.379	1.211	0.175
Baseline, black frames	0.441	1.408	0.246	0.528	1.602	0.300	0.476	1.568	0.287	0.318	1.052	0.150

Table 11: Turn response at step 38,400. Direction accuracy and ratios use left/right samples with $|y_{\mathrm{GT}}| > 0.5\,\mathrm{m}$.

	Baseline	Mask	No mask	Shuffled	Mirrored	No target	No img tokens	Black	Base. black
Corr. of final lateral offset	0.986	0.987	0.987	0.987	0.986	0.984	0.967	0.959	0.969
Std of predicted final lateral (m)	4.451	4.482	4.488	4.497	4.384	4.475	3.981	3.668	3.993
Left: direction accuracy	0.987	1.000	1.000	1.000	0.987	1.000	1.000	1.000	0.987
Right: direction accuracy	1.000	1.000	1.000	1.000	1.000	0.987	0.973	1.000	0.973
Left: median pred./GT lateral	0.993	1.002	1.000	0.993	0.961	0.984	0.919	0.897	0.931
Right: median pred./GT lateral	1.031	1.020	1.018	1.025	0.973	0.986	0.823	0.726	0.838

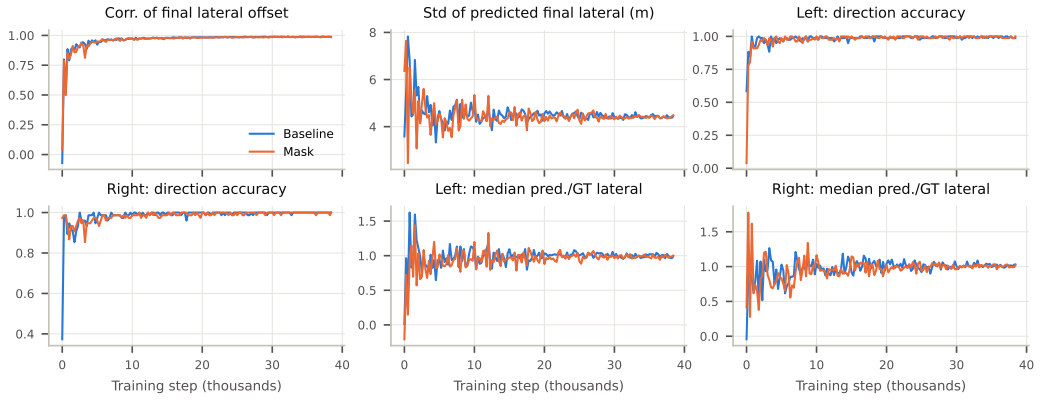


Figure 17: Turn response over training.

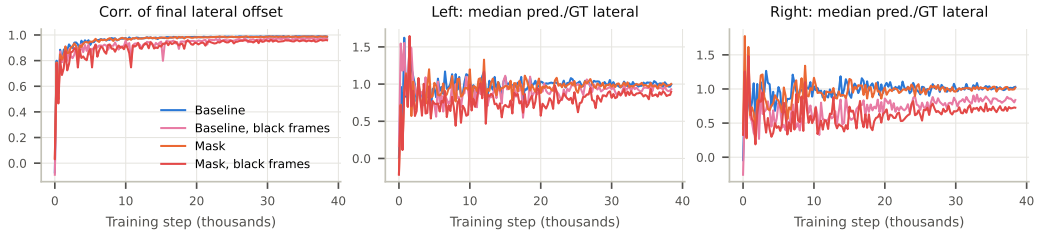


Figure 18: Turn response with the input frames replaced by black images.

Route. The challenge presets use MAP routes with a 40 m start offset. When the route's first point is farther than the resampling step, a cubic Hermite curve leaving the rig origin along $+x$ and joining the route's first segment bridges the gap; the route is then extended beyond the farthest target distance and resampled at 0.1 m. The target point is taken as in training, replacing the logged future with the route.

Serving. A plan is fixed in the local frame at creation. Each tick emits the plan's points at or after the current time on its own 10 Hz grid. If a plan cannot be made (missing camera or ego pose, or a

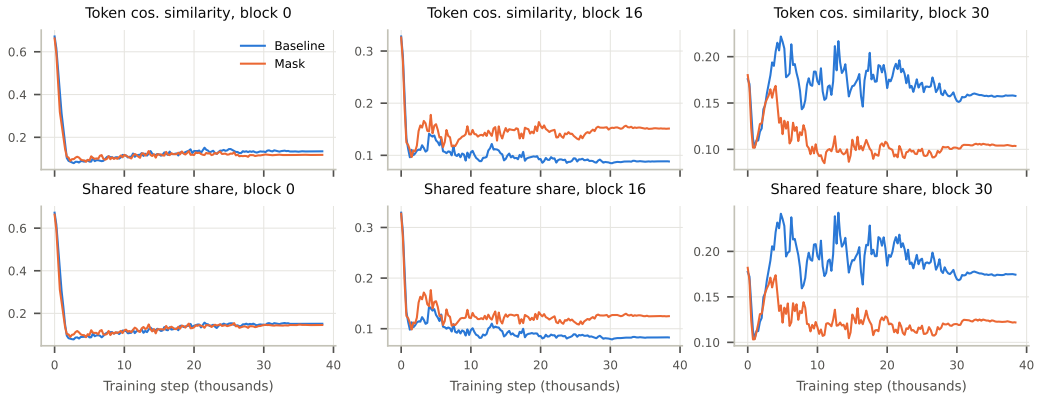


Figure 19: How similar the VLM tokens entering the DiT are, at three DiT blocks. In the deep block the mask run's tokens are less alike than the baseline's.

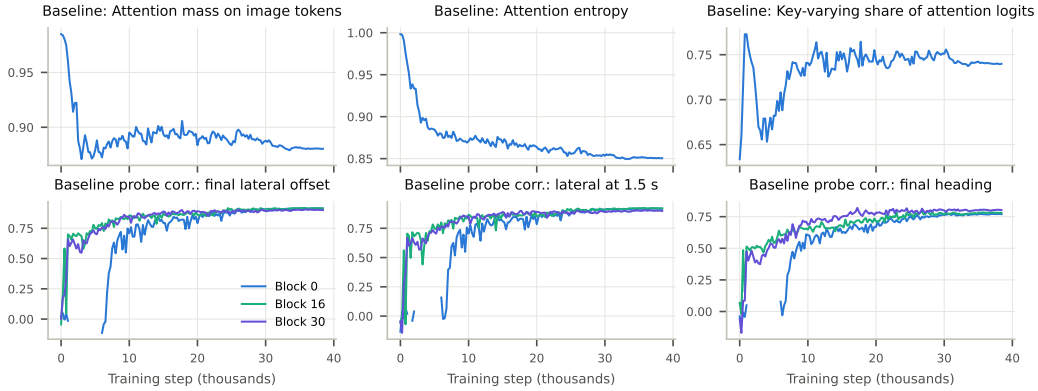


Figure 20: Baseline diagnostics: DiT attention statistics (top), and correlations of detached probes on projected VLM features at three blocks with future motion (bottom).

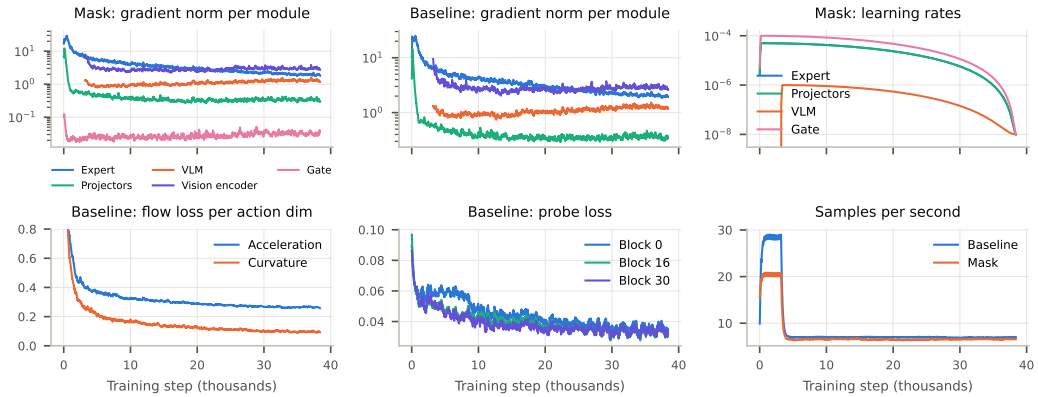


Figure 21: Optimization signals (smoothed). The first-epoch throughput is high because the VLM is frozen.

planning exception), the previous plan stays in use; with no usable plan, the driver serves a straight line at $\max(2\,\mathrm{m/s}, v)$.

AEB at a tick. (1) If the newest front frame has not been processed and is due, run the detector and project the boxes. (2) Transform the rest of the cached plan to the rig frame and find the lead along

it. (3) Update the track and compute the speed cap. (4) With a cap, let $w(\tau)$ be the distance the plan still means to cover in τ seconds, and $u(\tau)$ the distance allowed by the ramp (or $v\tau$ for the no-acceleration case). The served plan starts at the ego's projection onto the plan path, with arclength $s_{\text{ego}} + \min(w, u)$ on a 0.1 s grid and yaw interpolated along the path. (5) If the cap has just gone away, hold speed for this tick and drop the cached plan so that the next tick replans. Without step (5), the plan, now ahead of the braked ego, would make the controller accelerate toward what it braked for.

F AEB development

Why it was added. Without AEB, the largest single loss is at-fault collisions, mostly front hits, and most of them involve a vehicle well below the ego's speed. Across 178 first-incident front hits in five runs (mask, TE4, hist_ft, sig02), speed at impact has a median of 11.9 m/s (quartiles 6.8 and 17.8). The last 3 s are only moderately different from other 3 s windows before any incident (6,077 windows): speed gain above 2 m/s in 35% vs. 25%; mean acceleration above 0.3 m/s^2 in 48% vs. 37%; peak above 1.5 m/s^2 in 24% vs. 21%. A global acceleration limit would therefore mostly cost progress, so we chose a rule that looks at obstacles. The driver receives no other-actor state, and the recorded ground truth is off in the challenge presets, so obstacles must come from the cameras.

Offline replay. Each rollout log holds the front JPEGs, ego state and served plans. We ran the detector on them in the driver image and stored raw boxes, then evaluated the rule offline. A first version (no edge shrink, 1.6 m corridor, trigger 2 m/s^2) fired at least 1 s before 38 of 41 front hits, but also fired in 29 of 46 sampled clean rollouts. Inspection traced this mainly to parked cars at the kerb: a box's bottom corners lie beside an obliquely seen vehicle and intrude into the corridor. Table 12 shows the resulting trade-off. The replay is open loop and tuned on the evaluation scenes.

Table 12: Offline AEB threshold sweep on the replayed detections of the mask run's 150 rollouts (41 front hits, 62 rollouts without any incident). Brake counts only where the cap shortens the plan at 1 s by over 0.3 m. Chosen row in bold. Open loop: the ego does not react to the brake.

Shrink	Half width	Trigger	Gap	Samples	Hits, $\geq$ 1 s	Hits, $\geq$ 0.5 s	Clean: rollouts / frames
0.0	1.6	3.0	2.5	4	36	38	38 / 38
0.3	1.3	4.0	1.5	6	24	28	28 / 15
0.3	**1.3**	**3.0**	**2.5**	**6**	**29**	**32**	**32 / 19**
0.3	1.3	4.0	2.5	6	29	32	32 / 17
0.15	1.3	3.0	2.5	6	32	37	37 / 22
0.15	1.45	3.0	2.5	6	34	38	38 / 24
0.3	1.6	3.0	2.5	6	31	36	36 / 22
0.0	1.3	4.0	1.5	6	28	34	34 / 24
0.3	1.3	2.0	2.5	6	31	32	32 / 23

Chosen configuration. For the chosen row, the no-acceleration rule (3 s) adds 47 frames of intervention on clean rollouts. Rear-end collisions do not zero the score because only front and lateral collisions count as at fault, and progress saturates at 80% of the recorded distance, so the score rewards this trade-off. This does not make the trade-off safe on a real road (Section 5).

Detector choice. RT-DETRv2 is available in the image's `transformers` with Apache-2.0 weights, and DETR-style output needs no NMS code. R18 and R50 were timed in the image on a busy RTX PRO 6000 Blackwell: about 38 ms and 49 ms median per frame, with 183 and 298 MiB peak allocation. Resizing on the CPU and fp16 autocast did not change the R18 time (about 35 ms), and a batch of 4 took the same time as a batch of 1 (about 30 ms), so CPU-side kernel launches dominated under that host's load. CUDA-graph capture of the detector failed because the implementation writes to GPU tensors from the CPU in `forward`.

G Closed-loop results in full

Protocol. Scenes: a fixed 150-scene subset of the 441-scene `nurec_curated_val` suite. Preset `+e2e_challenge=dev`, with a multi-GPU topology and driver recording on, on hosts shared with other jobs. First rollouts (baseline, mask, 10 Hz AEB), TE4, hist_ft and 8-sample ran on an RTX PRO 6000 Blackwell host (8 GPUs). Second rollouts, sig02 and 5 Hz AEB ran on RTX 6000 Ada hosts (3 GPUs each). Every configuration therefore has its first and second rollout on different hardware, so the difference between a configuration's two rollouts mixes host and run-to-run variation.

Scenes lost to renderer out-of-memory errors on shared GPUs were re-run, with fewer concurrent rollouts per renderer. Rollouts stop being scored at the first collision or offroad event. A rollout's score is 0 on an at-fault collision (front or lateral), offroad, or lateral corridor exit (over 4 m from the recorded path), and otherwise $\min(\text{progress}/0.8, 1)$.

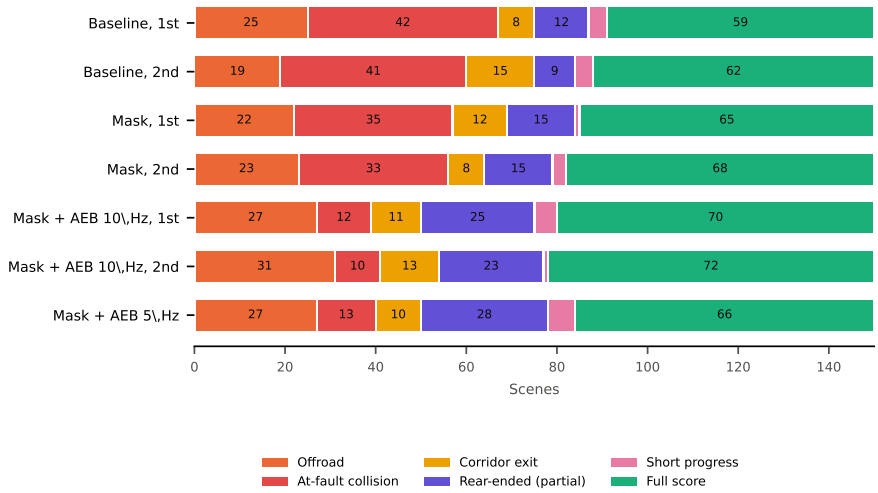


Figure 22: Outcome of each run on the same 150 scenes.

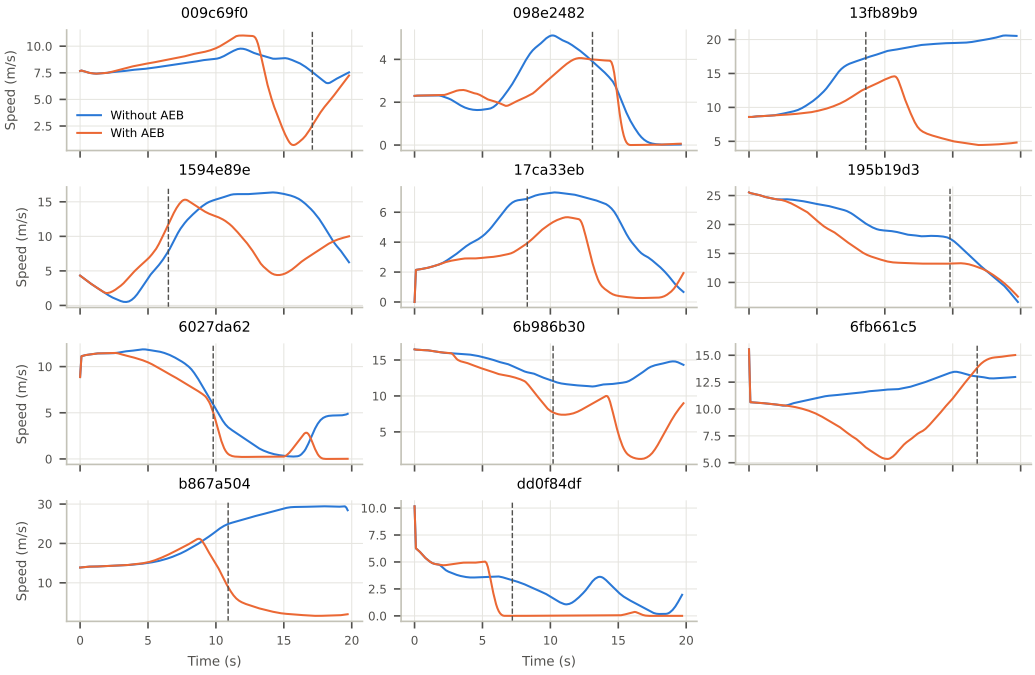


Figure 23: Ego speed in the 11 scenes with an at-fault collision without AEB and a full score with it (first rollouts). Dashed: the collision without AEB.

Collision kinematics without AEB. Table 16 compares models.

Remaining collisions with AEB (first rollout). Table 17 lists all 12. *Corridor*: the obstacle was outside the corridor until shortly before the hit or throughout; with no edge shrink and a 2 m half width, the same replay sees most of them earlier, at the price of the false brakes described above.

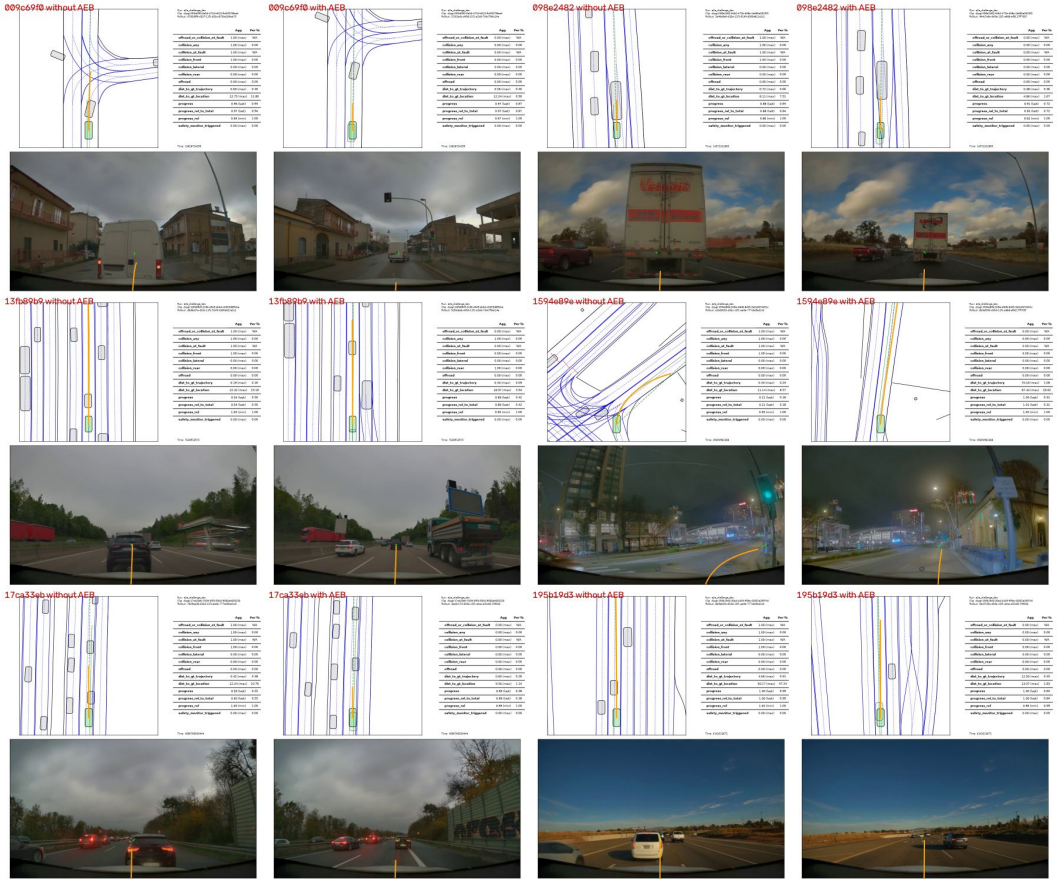


Figure 24: Same scene 0.3 s before the collision without AEB, at the same time from rollout start with AEB (6 of the 11 scenes).

Table 13: Score-loss decomposition on the same 150 scenes in every column: scenes / contribution to 1 − mean score.

	Baseline, 1st	Baseline, 2nd	Mask, 1st	Mask, 2nd	AEB 10 Hz, 1st	AEB 10 Hz, 2nd	AEB 5 Hz
Offroad	25 / 0.167	19 / 0.127	22 / 0.147	23 / 0.153	27 / 0.180	31 / 0.207	27 / 0.180
At-fault collision	42 / 0.280	41 / 0.273	35 / 0.233	33 / 0.220	12 / 0.080	10 / 0.067	13 / 0.087
Corridor exit	8 / 0.053	15 / 0.100	12 / 0.080	8 / 0.053	11 / 0.073	13 / 0.087	10 / 0.067
Rear-ended (partial)	12 / 0.022	9 / 0.015	15 / 0.038	15 / 0.037	25 / 0.054	23 / 0.046	28 / 0.067
Short progress	4 / 0.002	4 / 0.003	1 / 0.002	3 / 0.002	5 / 0.005	1 / 0.002	6 / 0.009
Full score	59 / 0.000	62 / 0.000	65 / 0.000	68 / 0.000	70 / 0.000	72 / 0.000	66 / 0.000
Mean score	0.476	0.481	0.499	0.534	0.607	0.591	0.590
Mean progress	0.745	0.773	0.737	0.739	0.737	0.732	0.720
Mean Drive (s)	0.133	0.066	0.110	0.146	0.201	0.085	0.264
Host	Blackwell	Ada	Blackwell	Ada	Blackwell	Ada	Ada

Side/behind: the actor approached from outside the corridor (crossing bus; vehicle from behind-left). *Close range*: the estimated gap stayed near 4.4 m, so no closing speed was seen; hood occlusion of the wheels is our hypothesis, not verified. *Late brake*: in `1316f5fa`, a distance jump restarted the track, so braking began about 0.5 s before the hit; in `382a7247`, the ego slowed from 9.2 to 0.7 m/s and still touched.

Lateral failures with AEB (first rollout). Of the 27 offroad scenes, 19 were also offroad in the first no-AEB rollout, and 3 had hard braking (below $-2\,\mathrm{m/s^2}$) in the last 3 s, so the AEB is not their main cause. 18 of 27 changed heading by less than 15° in the last 3 s, consistent with slow drift, and video of 12 sampled scenes often shows the plan heading to the right shoulder. Of 31 lateral failures with a measurable side, 17 are on the right and 9 on the left of the route, with 5 within 0.3 m.

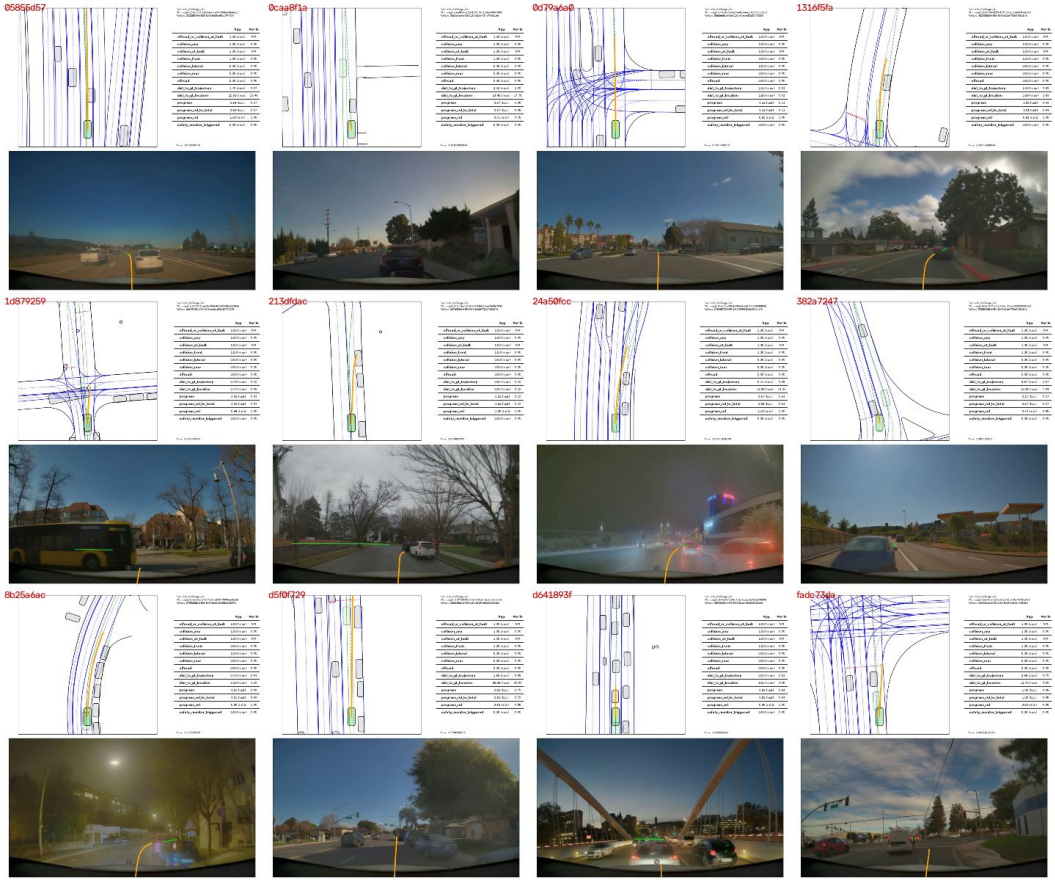


Figure 25: The 12 remaining at-fault collisions with AEB, 1 s before the collision.

Table 14: Paired per-scene comparisons on the same 150 scenes. CI: bootstrap with 10,000 resamples; p: two-sided sign test ignoring ties. Not corrected for multiple comparisons. (2): per-scene mean of two rollouts.

Comparison	n	Ref.	Run	Diff.	95% CI	W/L	p
Mask (2) vs. baseline (2)	150	0.479	0.517	+0.038	[-0.018, +0.093]	37/26	0.21
Mask 1st vs. baseline 1st	150	0.476	0.499	+0.023	[-0.049, +0.093]	28/20	0.31
Mask 2nd vs. baseline 2nd	150	0.481	0.534	+0.053	[-0.022, +0.126]	31/19	0.12
Mask 2nd vs. mask 1st	150	0.499	0.534	+0.035	[-0.017, +0.089]	22/16	0.42
Baseline 2nd vs. baseline 1st	150	0.476	0.481	+0.005	[-0.065, +0.076]	25/20	0.55
AEB 10 Hz (2) vs. mask (2)	150	0.517	0.599	+0.082	[+0.023, +0.143]	41/28	0.15
AEB 10 Hz 1st vs. mask 1st	150	0.499	0.607	+0.108	[+0.038, +0.176]	37/14	$<$0.01
AEB 10 Hz 2nd vs. mask 2nd	150	0.534	0.591	+0.057	[-0.024, +0.137]	34/26	0.37
AEB 10 Hz 2nd vs. 1st	150	0.607	0.591	-0.016	[-0.087, +0.054]	24/30	0.50
AEB 5 Hz vs. mask (2)	150	0.517	0.590	+0.073	[+0.012, +0.134]	43/22	0.01
AEB 5 Hz vs. mask 2nd	150	0.534	0.590	+0.056	[-0.013, +0.125]	36/21	0.06
AEB 5 Hz vs. AEB 10 Hz (2)	150	0.599	0.590	-0.009	[-0.065, +0.044]	34/34	1.00

Table 15: How scenes change when the AEB is added (first rollouts, same host, 150 scenes): row is the outcome without AEB, column with AEB.

Without \ with	Offroad	Coll.	Exit	Rear	Short	Full
Offroad	19		1			2
At-fault collision	2	9	2	11	2	9
Corridor exit	3		5	1		3
Rear-ended (partial)	1		1	11	2	
Short progress					1	
Full score	2	3	2	2		56

Table 16: First-incident front collisions on the same 150 scenes: rate; share with speed gain above 2 m/s in the last 3 s; share whose last plan did not slow down (planned speed 2 s ahead $\geq v-0.3$ m/s); share with the nearest actor ahead below 1 m/s (approximate); median speed at impact (m/s).

	Hits	Rate	Accel.	No slowing	Stopped lead	v_{hit}
Baseline (2)	90/301	29.9%	42%	68%	39%	11.6
Mask (2)	77/300	25.7%	30%	64%	38%	10.9
Mask + AEB 10 Hz (2)	28/300	9.3%	32%	50%	61%	9.1
Mask + AEB 5 Hz	12/150	8.0%	17%	42%	67%	2.4
hist_ft	46/147	31.3%	46%	67%	39%	13.2
sig02	50/150	33.3%	32%	62%	36%	12.9
Mask + TE4	41/150	27.3%	29%	66%	34%	10.5
Baseline + TE4	34/148	23.0%	35%	62%	44%	9.9

Table 17: The 12 at-fault collisions with AEB (10 Hz). Time from rollout start; speeds 3 s before and at the hit; minimum acceleration in the last 3 s; nearest other actor at the hit (speed, longitudinal and lateral centre offset in the ego frame, relative heading; approximate, it may not be the actor hit); seconds before the hit that the rule's brake condition first holds in detector replay. Classes are our judgment.

Scene	Class	Type	t	v_{-3}	v	$a_{\min}$	v_o	Lon.	Lat.	Head.	Brake
`0caa8f1a`	Close range	front	9.4	0.1	1.2	-1.5	0.5	4.4	-1.4	-10	4.5
`d641893f`	Close range	front	15.1	0.9	4.2	0.8	12.0	-1.1	3.1	-179	none
`05855d57`	Corridor	front	11.5	30.3	32.0	-0.7	26.8	4.3	-0.0	2	0.3
`213dfdac`	Corridor	front	3.2	11.3	8.7	-1.5	0.1	2.2	-1.9	7	none
`24a50fcc`	Corridor	front	11.3	9.8	9.7	-0.7	7.1	1.9	-1.9	4	none
`8b25a6ac`	Corridor	side	10.6	9.9	9.9	-0.1	0.0	2.7	-2.1	-1	none
`d5f0f729`	Corridor	front	18.2	24.3	21.7	-1.5	0.1	4.2	-1.6	1	none
`fadc73da`	Corridor	front	17.1	1.4	6.5	-0.2	0.0	3.7	2.1	4	none
`1316f5fa`	Late brake	front	19.8	6.6	7.0	-8.3	0.1	4.3	-1.7	-5	0.5
`382a7247`	Late brake	front	9.0	9.2	0.7	-8.6	0.0	4.0	1.8	-162	2.9
`0d79a6a0`	Side/behind	side	3.2	10.5	8.1	-5.5	11.8	0.1	2.5	-22	0.9
`1d879259`	Side/behind	front	6.0	2.1	4.6	-2.9	6.0	4.6	-0.8	-71	0.3

Route as a lateral reference. Routes are generated from map lanes, so they are a candidate lateral reference. Sampling every 2 m and placing the ego footprint along the route, 114 of 36,843 poses (0.3%) count as offroad under the scorer's logic (5,728 do not fit inside one lane but touch no road edge). So the route is almost always on the road, but Tables 18 and 19 show that distance to it separates failures from normal driving poorly, so we did not deploy a guard.

Table 18: Ego distance to the served route before lateral failures with AEB (m). Scenes without a covering route segment are omitted. For comparison, frames of the 70 full-score rollouts: median 0.29, 90th percentile 1.14, 99th percentile 2.99.

Failure	Before	n	Median	p25	p75	>1 m	>2 m
Offroad	-3 s	18	0.40	0.18	2.56	5	5
	-2 s	23	0.49	0.31	1.36	6	4
	-1 s	23	0.70	0.33	1.91	11	5
	-0 s	23	0.67	0.40	1.99	8	6
Corridor exit	-3 s	5	0.81	0.70	0.87	1	1
	-2 s	7	0.79	0.34	2.77	3	3
	-1 s	8	0.82	0.26	2.36	4	3
	-0 s	8	2.80	0.61	3.43	5	5

Table 19: Offline check of a route-distance guard (not deployed): lateral failures flagged at least 1 s before the failure, and full-score rollouts ever flagged.

Threshold (m)	Ego: failures	Ego: clean	Plan at 2 s: failures	Plan at 2 s: clean
0.5	27/38	50/70	29/38	49/70
0.75	21/38	37/70	26/38	39/70
1.0	19/38	30/70	23/38	33/70
1.5	15/38	17/70	17/38	20/70
2.0	13/38	15/70	15/38	16/70
3.0	8/38	7/70	12/38	9/70

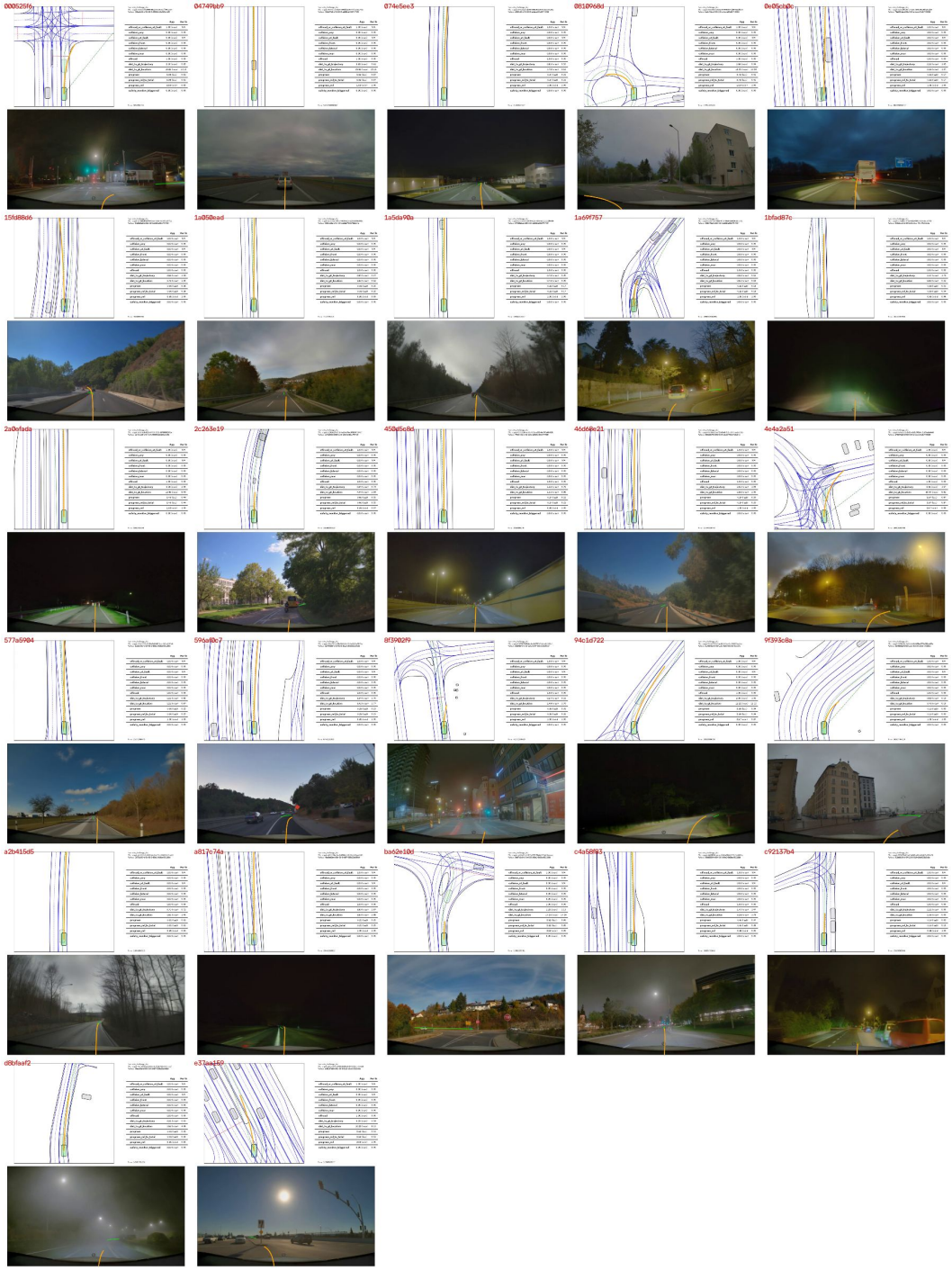


Figure 26: All offroad failures with AEB (first rollout), 0.8 s before the event.

H Variants evaluated and not submitted

All use the same local protocol and no AEB. These are reported as negative or inconclusive results.

- **Temporal ensembling (TE4).** In the spirit of ACT [3]: each new plan is averaged with up to 3 older plans where they overlap in time, each rigidly re-anchored so its pose at the new plan's creation time equals the current pose, with weights $\exp(-0.5 \cdot \text{age})$.

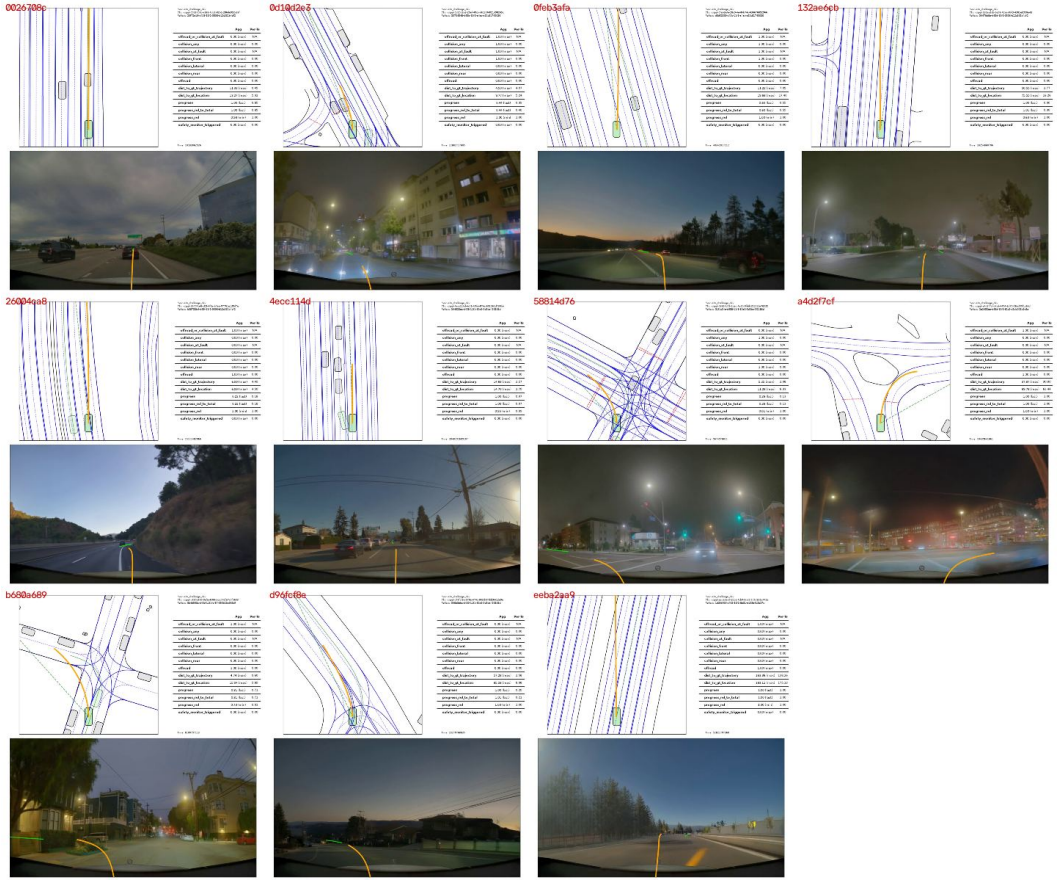


Figure 27: All lateral corridor exits with AEB (first rollout), 0.8 s before the incident or end of scoring.

- **8 samples.** 8 flow-matching samples per plan averaged in the normalized action space.
- **hist_ft.** 3-epoch mask fine-tune from the baseline's final checkpoint: ego-history dropout $p = 0.5$ with a presence flag (past x, y, heading, speed and all a_{lon} zeroed; the current speed is always kept), past-motion dropout after ChauffeurNet [2], a dropped target where it is a clip's fallback endpoint, and 300 gate-only steps.
- **sig02.** As hist_ft with a sigmoid mask (initial keep 0.2) and a budget penalty of weight 0.1.

Both fine-tunes trained for only 3 epochs.

Table 20: Variants we evaluated and did not submit (no AEB), paired as in Table 14, one rollout each against the mean of two.

Comparison	n	Ref.	Run	Diff.	95% CI	W/L	p
Baseline + TE4 vs. baseline (2)	150	0.479	0.504	+0.026	[-0.027, +0.079]	33/22	0.18
Mask + TE4 vs. mask (2)	150	0.517	0.495	-0.021	[-0.082, +0.041]	25/29	0.68
Mask + 8 samples vs. mask (2)	150	0.517	0.512	-0.005	[-0.051, +0.040]	24/21	0.77
hist_ft vs. mask (2)	150	0.517	0.495	-0.021	[-0.082, +0.037]	29/26	0.79
sig02, 1st vs. mask (2)	150	0.517	0.505	-0.012	[-0.077, +0.055]	27/28	1.00
sig02, 2nd vs. mask (2)	150	0.517	0.497	-0.020	[-0.084, +0.043]	29/29	1.00

I Runtime measurements

All local measurements share hosts with other jobs and differ from the official evaluator, which runs 16 replicas on 4 GPUs with 2 rollouts per replica. Official numbers appear in submission status. Table 13 gives the driver telemetry of evaluation runs (recording on, up to 12 rollouts per driver).

Table 21: Baked submission images, started without arguments in the sandbox: 2 replicas with 1 session each on one GPU shared with another job, about 27 s. Times in s.

	Mean Drive	Planning call	Other calls, mean	Other calls, p50	Calls with detector
AEB 10 Hz	0.111	0.249	0.077	0.026	143/143
AEB 5 Hz	0.104	0.230	0.072	0.015	82/143

Table 22: Saturation test: recorded rollouts replayed into 3 replicas (submission sandbox, no recording) on one RTX 6000 Ada also used by another job, 2 sessions each, with no simulator between calls. This overstates contention; compare rows only. Times in s.

	Sessions	Ticks	Mean	p50	p90	p99	Per tick	Wall
No AEB	6	1071	0.294	0.001	1.457	1.512	0.322	64
AEB 10 Hz	6	1071	0.317	0.169	0.890	1.125	0.346	71
AEB 5 Hz	6	1071	0.306	0.113	1.167	1.365	0.333	67

In the saturation test, the mean Drive increases by 8% at 10 Hz and 4% at 5 Hz relative to no AEB. The absolute values are not model latency: 6 sessions compete for one GPU with no idle time between calls, in contrast with a simulator that spends time rendering.

J Sources of the reported data

Open-loop metrics, training signals and Figs. 3–11 come from W&B runs `ccb412c9` and `03bdfc50` in project `starVLA_drive`, both started from commit `b5c4446` with seed 42. Closed-loop data come from each run's `aggregate/results-summary.json`, per-step `metrics.parquet` and `rollout.asl`. Deployed model code differs from the training commit only by options that are off for this checkpoint (history dropout and fallback-target handling).